Wavelet-Based Time-Frequency Analysis of Evolving fluids in a Microfluidic Rheometer

Subject:	Meeting with Supervisor | Draft for review
Group name/ identifier	Group 24
Meeting Location, Date & Time	Location: Not recorded Date: 20 August 2026 Time: Not recorded
Group members present	Rohan Mathew, Jianhao Zhang, Xuan Zhang, Ria Mishra, Xingcan Wang (Alex). Supervisor: Saumya.
Apologies	None

1 Actions from previous meetings

Task	**Who is responsible**	**Deadline**
Investigate incremental ablation studies to assess individual feature contributions.	All members	Follow-up from 13 August
Apply relevant parameter tuning and step response techniques to the July 18 dataset.	Jianhao	Follow-up from 13 August
Prepare functional Python notebooks and quantitative results.	All members	Follow-up from 13 August

2 Items discussed and decisions made

Item	**Decision (if applicable)**
Pipeline refinement	Proposed follow-up: refine the data loading, protocol classification, quality screening and method comparison workflow demonstrated on 13 August. Retain consistent inputs when comparing methods.
Feature contribution and fair comparison	Review how small ablation studies could separate the contribution of wavelets, harmonic distortion and other features. Use comparable error measures and document the conditions under which each method is useful.
Physical interpretation of detected changes	Distinguish a detected structural change from a validated gelation event. Review storage/loss moduli, loss tangent, phase angle and frequency dependence as possible supporting criteria.
Ground truth and synthetic signals	The available real data do not provide an exact clotting reference for every recording. Synthetic signals can support controlled evaluation, but their physical assumptions and limits must be justified from the literature.
Physics informed analysis	Continue investigating how mechanistic knowledge or a physically meaningful derived signal could complement wavelet analysis. Treat potential model extensions as exploratory until their suitability for blood data is established.
Preparation for advisor review	Organize code, figures and open questions for the 27 August discussion. Prioritize clearer gelation criteria, evidence of structural change and the limitations of the available recordings.

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3 Agreed actions from this meeting

Task	**Who is responsible**	**Deadline**
Refine the analysis notebook and prepare code and figures for advisor review.	Rohan	For review on 27 August
Continue literature review on physics based models and defensible gelation criteria.	All members	For review on 27 August
Assess available moduli, loss tangent and frequency sweep information without forcing a gel point label.	All members	For review on 27 August
Document data quality limitations and distinguish synthetic benchmarks from real blood evidence.	All members	For review on 27 August
Prepare questions about linking detected structural changes to physical behaviour.	All members	For review on 27 August

4 Next meeting

Location	Not recorded
Date & Time	27 August 2026; time to be confirmed
Goals	Review updated code, rheological criteria and potential physics informed extensions with the supervisor.

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